

Subject: All of Statistics, Random Variables

Date: from May 29, 2026 to June 2, 2026

Contents

1	Expectation	1
---	-------------	---

1 Expectation

Expectation

Expectation. 随机变量 X 的期望/一阶矩 (expectation, mean, first moment) 定义为

$$\mu_X = \mathbb{E}(X) = \int x dF(x) = \begin{cases} \sum_x x f(x) & \text{if } X \text{ is discrete} \\ \int x f(x) dx & \text{if } X \text{ is continuous} \end{cases} \quad (1)$$

其存在条件为 $\int_x |x| dF_X(x) < \infty$.

对于变换 $Y = r(X)$, 其期望为^a:

$$\mathbb{E}(Y) = \mathbb{E}(r(X)) = \int r(x) dF_X(x). \quad (2)$$

因此概率也是一种特殊情形的期望: $\mathbb{E}(I_A(X)) = \int I_A(x) f_X(x) dx = \int_A f_X(x) dx = \mathbb{P}(X \in A)$.

对于变换 $Z = r(X, Y)$, 其期望为:

$$\mathbb{E}(Z) = \mathbb{E}(r(X, Y)) = \int \int r(x, y) dF(x, y). \quad (3)$$

Moment. 随机变量的 k 阶原点矩定义为 $\mathbb{E}(X^k) = \int_{-\infty}^{\infty} x^k dF(x)$, k 阶中心矩 (Central Moment) 定义为 $\mathbb{E}[(X - \mathbb{E}(X))^k]$. 若 k 阶矩 (k^{th} moment) 存在, 则 $\forall j < k$, j 阶矩也存在.

Property of Expectation. ① 若 $X_1, \dots, X_n$ 为随机变量, $a_1, \dots, a_n$ 为常数, 则 $\mathbb{E}(\sum_i a_i X_i) = \sum_i a_i \mathbb{E}(X_i)$. ② 若 $X_1, \dots, X_n$ 为 i.i.d. 的随机变量, 则 $\mathbb{E}(\prod_{i=1}^n X_i) = \prod_i \mathbb{E}(X_i)$.

^a这被称为 Law of the Unconscious Statistician (LOTUS). 严格来说 $\mathbb{E}(Y) = \int_{-\infty}^{\infty} y dF_Y(y)$, 但现在只需要把 $r(x)$ 塞进 X 的积分中即可. 这本质是测度论中测度推进 (Push-forward Measure) 下的变量替换定理 (Change of Variables).

Variance and Covariance

Variance. 方差衡量了分布的“离散程度”, 定义为:

$$\text{Var}(X) = \sigma^2 = \mathbb{E}(X - \mu)^2 = \int (x - \mu)^2 dF(x). \quad (4)$$

标准差则是方差的平方根, 为 $\text{std}(X) = \sqrt{\text{Var}(X)} = \sigma$.

设 $X_1, \dots, X_n$ 为随机变量, 定义样本均值 (sample mean), 样本方差 (sample variance) 分别为

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i, \quad S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2. \quad (5)$$

若 $X_1, \dots, X_n$ i.i.d., 则 ① $\mathbb{E}(\bar{X}_n) = \mu$ ② $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$ ③ $\mathbb{E}(S_n^2) = \sigma^2$.

Proof. ① $\mathbb{E}(\bar{X}_n) = \mathbb{E}(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{1}{n} \sum_{i=1}^n \mathbb{E}(X_i) = \frac{1}{n} \cdot (n\mu) = \mu$. ② $\text{Var}(\bar{X}_n) = \text{Var}(\frac{1}{n} \sum_{i=1}^n X_i) = \frac{1}{n^2} \text{Var}(\sum_{i=1}^n X_i) = \frac{1}{n^2} \cdot (n\sigma^2) = \frac{\sigma^2}{n}$. ③ $\sum_{i=1}^n (X_i - \bar{X}_n)^2 = \sum_{i=1}^n (X_i^2 - 2X_i \bar{X}_n + \bar{X}_n^2) = \sum_{i=1}^n X_i^2 - 2\bar{X}_n \sum_{i=1}^n X_i + n\bar{X}_n^2 = \sum_{i=1}^n X_i^2 - n\bar{X}_n^2$. 因此 $\mathbb{E}[\sum_{i=1}^n (X_i - \bar{X}_n)^2] = \sum_{i=1}^n \mathbb{E}(X_i^2) - n\mathbb{E}(\bar{X}_n^2) = \sum_{i=1}^n (\sigma^2 + \mu^2) - n(\frac{\sigma^2}{n} + \mu^2) = (n-1)\sigma^2$. 故有 $\mathbb{E}(S_n^2) = \frac{1}{n-1} \cdot (n-1)\sigma^2 = \sigma^2$. $\square$

Property of Variance. ① $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$. ② $\text{Var}(aX + b) = a^2\text{Var}(X)$, 其中 a 为常数. ③ 若 $X_1, \dots, X_n$ 相互独立, $a_1, \dots, a_n$ 为常数, 则 $\text{Var}(\sum_{i=1}^n a_i X_i) = \sum_{i=1}^n a_i^2 \text{Var}(X_i)$.

Covariance. 协方差衡量了随机变量 X, Y 之间线性关系的强度, 其定义为:

$$\text{Cov}(X, Y) = \mathbb{E}\left((X - \mu_X)(Y - \mu_Y)\right), \quad (6)$$

并定义 X, Y 的 correlation 为 $\rho = \rho_{X,Y} = \rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$.

Property of Covariance. ① $\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$. ② $-1 \leq \rho(X, Y) \leq 1$. ③ 若 X, Y 满足线性关系 $Y = aX + b$, 则 $\rho(X, Y) = 1, a > 0, \rho(X, Y) = -1, a < 0$. 若 X, Y 相互独立则 $\rho(X, Y) = 0$. ④ $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$, $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)$, $\text{Var}(\sum_i a_i X_i) = \sum_i a_i^2 \text{Var}(X_i) + 2 \sum \sum_{i < j} a_i a_j \text{Cov}(X_i, X_j)$.

Distribution	PMF / PDF $f(x)$	Mean $\mathbb{E}(X)$	Variance $\text{Var}(X)$
Point mass at a	$P(X = a) = 1, P(X \neq a) = 0$	a	0
Bernoulli(p)	$p^x(1-p)^{1-x}, \quad x \in \{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$\binom{n}{x} p^x (1-p)^{n-x}, \quad x \in \{0, \dots, n\}$	np	$np(1-p)$
Geometric(p)	$p(1-p)^{x-1}, \quad x \in \{1, 2, \dots\}$	$1/p$	$(1-p)/p^2$
Poisson(λ)	$\frac{e^{-\lambda} \lambda^x}{x!}, \quad x \in \{0, 1, \dots\}$	λ	λ
Uniform(a, b)	$\frac{1}{b-a}, \quad x \in [a, b]$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$	μ	σ^2
Exponential(β)	$\frac{1}{\beta} e^{-x/\beta}, \quad x \geq 0$	β	β^2
Gamma(α, β)	$\frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, \quad x > 0$	$\alpha\beta$	$\alpha\beta^2$
Beta(α, β)	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad x \in (0, 1)$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$
t_ν	$\frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$	$0 \quad (\nu > 1)$	$\frac{\nu}{\nu-2} \quad (\nu > 2)$
χ_p^2	$\frac{1}{2^{p/2}\Gamma(p/2)} x^{p/2-1} e^{-x/2}, \quad x > 0$	p	$2p$
Multinomial(n, p)	$\frac{n!}{\prod_i x_i!} \prod_i p_i^{x_i}, \quad \sum_i x_i = n$	np_i	$\begin{cases} np_i(1-p_i) & \text{if } i = j \\ -np_i p_j & \text{if } i \neq j \end{cases}$
Multivariate Normal	$\frac{1}{(2\pi)^{d/2} \Sigma ^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right)$	μ	Σ

Conditional Expectation

Conditional Expectation. 给定随机变量 X, Y , 则已知 $Y = y$ 条件下 X 的条件期望为:

$$\mathbb{E}(X|Y = y) = \begin{cases} \sum x f_{X|Y}(x|y) dx & \text{discrete case} \\ \int x f_{X|Y}(x|y) dx & \text{continuous case.} \end{cases} \quad (7)$$

对于 $r(x, y)$, 有

$$\mathbb{E}(r(X, Y)|Y = y) = \begin{cases} \sum r(x, y) f_{X|Y}(x|y) dx & \text{discrete case} \\ \int r(x, y) f_{X|Y}(x|y) dx & \text{continuous case.} \end{cases} \quad (8)$$

注意: $\mathbb{E}(X)$ 为一个值, 但未知 $Y = y$ 的前提下 $\mathbb{E}(X|Y)$ 仍然是一个随机变量.

Conditional Variance. 给定随机变量 X, Y , 则已知 $Y = y$ 条件下 X 的条件方差为: $\text{Var}(Y|X = x) = \int (y - \mu(x))^2 f(y|x) dy$, 其中 $\mu(x) = \mathbb{E}(Y|X = x)$.

Property. ① (Law of Total Expectation/Tower Property) $\mathbb{E}[\mathbb{E}(Y|X)] = \mathbb{E}(Y)$, $\mathbb{E}[\mathbb{E}(X|Y)] = \mathbb{E}(X)$, $\mathbb{E}[\mathbb{E}(r(X, Y)|X)] = \mathbb{E}(r(X, Y))$. ② (Law of Total Variance) $\text{Var}(Y) = \mathbb{E}[\text{Var}(Y|X)] + \text{Var}[\mathbb{E}(Y|X)]$. 其中 $\mathbb{E}[\text{Var}(Y|X)]$ 为组内方差 (within variance), $\text{Var}[\mathbb{E}(Y|X)]$ 为组间方差 (between variance).

Proof. ① $\mathbb{E}[\mathbb{E}(Y|X)] = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} y f(x|y) dy \right) f_X(x) dx = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f(x, y) dy dx \xrightarrow{\text{Fubini's Theorem}} \int_{-\infty}^{\infty} y \left(\int_{-\infty}^{\infty} f(x, y) dx \right) dy = \int_{-\infty}^{\infty} y f_Y(y) dy = \mathbb{E}(Y)$. ② 根据 $\text{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2$, 我们有 $\mathbb{E}[\text{Var}(Y|X)] = \mathbb{E}[\mathbb{E}(Y^2|X) - [\mathbb{E}(Y|X)]^2] = \mathbb{E}(Y^2) - \mathbb{E}[\mathbb{E}(Y|X)]^2$, $\text{Var}[\mathbb{E}(Y|X)] = \mathbb{E}[\mathbb{E}(Y|X)]^2 - (\mathbb{E}[\mathbb{E}(Y|X)])^2 = \mathbb{E}[\mathbb{E}(Y|X)]^2 - [\mathbb{E}(Y)]^2$. 故 $\mathbb{E}[\text{Var}(Y|X)] + \text{Var}[\mathbb{E}(Y|X)] = \mathbb{E}(Y^2) - [\mathbb{E}(Y)]^2 = \text{Var}(Y)$. $\square$

Moment Generating Functions

Moment Generating Functions (MGF). 定义随机变量 X 的矩生成函数 (或 Laplace transform) 为

$$M_X(t) = \mathbb{E}(e^{tX}) = \begin{cases} \sum_x e^{tx} p(x) & (\text{若 } X \text{ 离散}) \\ \int e^{tx} dF(x) & (\text{若 } X \text{ 连续}), \end{cases} \quad (9)$$

其中 $t \in \mathbb{R}$. $M_X(t)$ 存在条件^a为: 存在一个正数 $h > 0$, 使得当 $t \in (-h, h)$ 时, $M_X(t) < \infty$.

Property. ① 根据 Taylor 展开可以得到

$$M_X(t) = \mathbb{E}(e^{tX}) = 1 + t\mathbb{E}(X) + \frac{t^2}{2!}\mathbb{E}(X^2) + \frac{t^3}{3!}\mathbb{E}(X^3) + \cdots + \frac{t^k}{k!}\mathbb{E}(X^k) + \cdots$$

因此 $\frac{d^k}{dt^k} M_X(t) \Big|_{t=0} = \mathbb{E}(X^k)$. ② 若 $Y = aX + b$, 则 $M_Y(t) = e^{bt} M_X(at)$.

③ 若 $X_1, \dots, X_n$ 相互独立, $Y = \sum_{i=1}^n X_i$, 则 $M_Y(t) = \prod_{i=1}^n M_{X_i}(t)$.

④ 若两个随机变量 X 和 Y 在 $t = 0$ 的某个邻域内 $M_X(t) = M_Y(t)$, 那么 X 和 Y 必然具有完全相同的概率分布 ($X \stackrel{d}{=} Y$).

^a即并非除了 $t = 0$ 之外积分都发散.

Distribution	Moment Generating Function $M_X(t) = \mathbb{E}(e^{tX})$	Condition / Domain
Point mass at a	e^{at}	$\forall t \in \mathbb{R}$
Bernoulli(p)	$(1 - p) + pe^t$	$\forall t \in \mathbb{R}$
Binomial(n, p)	$((1 - p) + pe^t)^n$	$\forall t \in \mathbb{R}$
Geometric(p)	$\frac{pe^t}{1 - (1-p)e^t}$	$t < -\ln(1 - p)$
Poisson(λ)	$\exp(\lambda(e^t - 1))$	$\forall t \in \mathbb{R}$
Uniform(a, b)	$\begin{cases} \frac{e^{tb} - e^{ta}}{t(b-a)} & \text{if } t \neq 0 \\ 1 & \text{if } t = 0 \end{cases}$	$\forall t \in \mathbb{R}$
Normal(μ, σ^2)	$\exp(\mu t + \frac{1}{2}\sigma^2 t^2)$	$\forall t \in \mathbb{R}$
Exponential(β)	$(1 - \beta t)^{-1}$	$t < 1/\beta$
Gamma(α, β)	$(1 - \beta t)^{-\alpha}$	$t < 1/\beta$
χ_p^2	$(1 - 2t)^{-p/2}$	$t < 1/2$

Last updated: June 2, 2026